Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 1 – FUELS, GASES & THERMOCHEMISTRY

Part B

MARKING

Date: Tuesday, March 10th

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	40	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided and you may use your Unit 3 SAC 1 logbook from Part A
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task...

- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.

Electronic Devices

Students are <u>not</u> allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Section A (12 marks)

Q1. Summary of analysis (refer to logbook for working) (6 marks)

formula of alcohol	
mass of fuel burned (g)	
amount, in mol, of fuel burned (mol)	
Energy, q, absorbed by water (J)	
Energy, q, released by combustion of the alcohol (J)	
Energy released per mol of fuel during combustion (molar heat of combustion) (kJ mol^{-1})	conseq. mark possible.

* each.
- all consistent with Logbook results.

(-1) once if negative values given.

Q2. Based on your results, write a thermochemical equation for the complete combustion of the alcohol.
- fractional coefficients OK.

(3 marks)

* formulae ½ states ½ (fuel(l), H_2O(l/g))
* balance
* ΔH - value from ↓
  - must have sign (-)
  - must have unit

Q3. Discuss the reliability of your data and suggest **two** ways in which the experimental design might be improved.

(3 marks)

(1) value not reliable OR reliability cannot be assessed

(1) repeat and average (multiple trials)

(1) one other change that minimises random error
eg shield flame to prevent flame dancing, use bomb calorimeter, others.

Possible answers to Q2: (X = value from last row in Q1)

methanol: $CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l/g)$ $\Delta H = -X$ kJ mol^{-1}
OR $2CH_3OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 4H_2O(l/g)$ $\Delta H = -2X$ kJ mol^{-1}

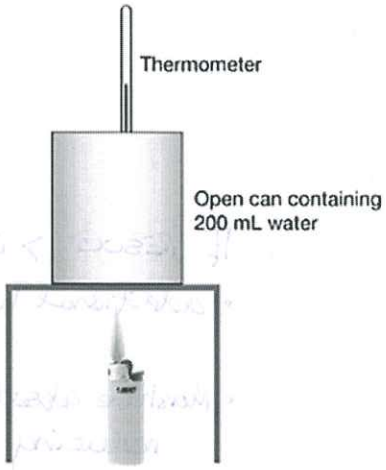
ethanol: $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l/g)$ $\Delta H = -X$ kJ mol^{-1}

propan-2-ol: $C_3H_7OH(l) + \frac{9}{2}O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l/g)$ $\Delta H = -X$ kJ mol^{-1}
OR 2 " 9 " 6 " 8 " $\Delta H = -2X$ kJ mol^{-1}

Section B

Q4. *(10 marks)*

Butane, C_4H_{10}, is a liquid hydrocarbon that is used as a fuel. A student designed an experiment similar to yours to measure the heat of combustion of butane, as shown in the diagram.



Mass of water in can = 200 g
Mass of butane burnt = 1.50 g
Initial temperature of water = 18.2°C
Final temperature of water = 63.4°C

a. Use the temperature readings provided to calculate the temperature change (ΔT) in the water in the can. (1 mark)

45.2 °C (or K).
(½) (½)

b. Use the specific heat capacity of water (from the data book) and ΔT to determine the amount of energy that has been transferred to the water. (1 mark)

$q = 200 \times 4.18 \times 45.2$
$= 3.78 \times 10^4$ J
$= 37.8$ kJ
(3sf).
either

½ if 0.200 (kg) used
– out by 10^3

c. Write a balanced chemical equation for the complete combustion of butane, C_4H_{10}. (2 marks)

① $C_4H_{10}(l/g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l/g)$
② OR 2 " 13 " 8 " 10 "

* formulae/states
* coefficients/balan

d. Calculate the amount, in mole, of butane that underwent combustion during the experiment. (1 mark)

$n(C_4H_{10}) = \frac{1.50}{58.0} = 0.0259$ * mol. (unit not required).
(2.59×10^{-2})

e. Use your answers to calculate ΔH for the combustion reaction of butane according to the equation given in question (c). (3 marks)

Energy per mole $= \frac{37.8 \text{ (conseq)}}{0.0259 \text{ (conseq.)}}$

$= 1.46 \times 10^3$ kJ mol. (conseq) ⇒ for ① $\Delta H = -1.46 \times 10^3$ kJ mol^{-1}
② $\Delta H = -2.92 \times 10^3$ kJ mol^{-1}
* * (conseq) *

f. By comparing your results with the literature molar heats of combustion given below, give two sources of experimental error that can account for the discrepancy between the experimental result obtained here and the value provided in the data book. (2 marks)

* each of 2
– must be consistent with observed discrepancy

Alcohol	Molar heat of combustion (kJ mol^{-1})
methanol	726
ethanol	1360
propanol	2016

If RESULT < EXPECTED
- some heat absorbed by container
- some heat lost to surroundings
- incomplete combustion
- some fuel evaporated.

If RESULT > EXPECTED
- additional heat from burning wick.
- moisture absorbed by fuel, reducing size of mass decrease.

Q5. *(8 marks)*
Butane is a component of liquefied petroleum gas (LPG) which is used to power some cars. A tank of butane can supply 2500 MJ of energy for driving.

a. Use Table 11 in the data book to determine the mass of butane needed to supply this much energy. (2 marks)

Heat of combustion = 49.7 kJ g^{-1}

$m = \frac{2500 \times 10^3}{49.7} = 50.3$ kg (or equiv.) *

3 sig fig *

b. Use your answer from 5a to determine the mass of carbon dioxide emitted into the atmosphere during the combustion of this mass of butane. (3 marks)

$n(C_4H_{10}) = \frac{50.3 \times 10^3 \text{ (consq.)}}{58.0} = 867$ mol *

$n(CO_2) = \frac{4}{1} \times n(C_4H_{10}) = 3.47 \times 10^3$ mol *

$m(CO_2) = n(CO_2) \times 44.0 = 1.53 \times 10^5$ g = 153 kg * (or equiv)

c. In another experiment, 2100 MJ of energy was produced by a tank of butane. This produced 128 kg of CO_2 gas. Determine the mass of CO_2 emitted per MJ of energy released from combustion of butane. (1 mark)

mass CO_2 per MJ = $\frac{128 \times 10^3}{2100} = 61.0$ g *

d. Biogas, which is predominantly made of methane, is used as a fuel. The equation for complete combustion of methane is:

$$CH_4\,(g) + 2O_2\,(g) \rightarrow CO_2\,(g) + 2H_2O\,(g)$$

The combustion of enough methane to give 3000 MJ produces 148.3 kg of CO_2. Determine the volume, in litres, that this amount of CO_2 would occupy at standard laboratory conditions (SLC). (2 marks)

$n(CO_2) = \frac{148.3 \times 10^3}{44.0} = 3370$ mol *

$V(CO_2) = nV_m = 3370 \times 24.8 = 8.36 \times 10^4$ L *

(unit not req^d if correct in L)

Q6. (4 marks)

Identify the advantages and disadvantages of using natural gas and bioethanol for fuel by comparing the two in terms of their energy content, net effects on atmospheric carbon dioxide levels, renewability and sustainability. for each category * ½ for clear statement of comparison. * ½ for some valid point of explanation/justification.

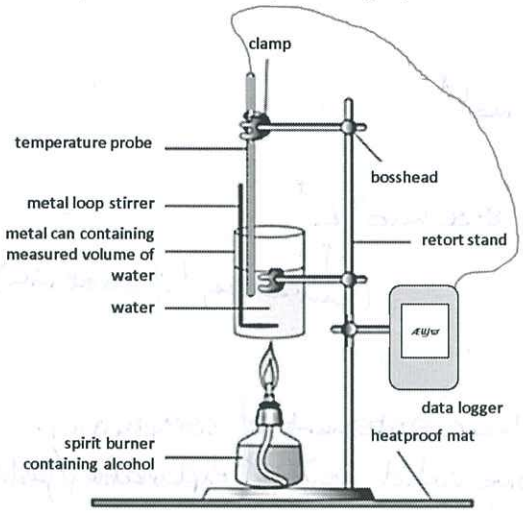
Fuel Type	Energy content (Use databook values)	Net CO_2 levels	Renewability	Sustainability
Bioethanol	29.6 kJ g^{-1}	- closer to carbon-neutral - CO_2 absorbed during plant growth.	- Renewable. - derived from sources that can be re-grown	More sustainable eg • renewable • can used recycled wastes. but • limited by processing costs • limited by competition for arable land
Natural gas	54.0 kJ g^{-1}.	- net CO_2 increase from combustion	Non-renewable. - limited reserves.	less sustainable • non-renewable • net CO_2 increase so climate change contributor

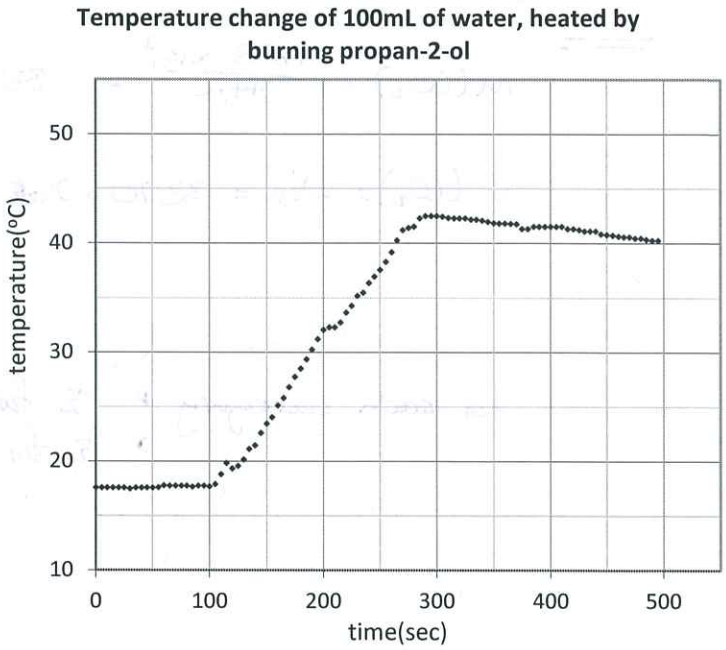
½* kJ g^{-1} values
½* comment for comparison.

- deduct ½ for any inaccurate/incorrect statements.

Q7. *(6 marks)*

Another student used the same apparatus as you have used but substituted the thermometer with a temperature probe for more accurate recordings with the same aim as the experiment you conducted. The burner was lit after 100 seconds and extinguished at 270 seconds. The fuel used was propan-2-ol. Below is their experimental setup and a graph of their data.





Mass of water in can = 100 g
Mass of propan-2-ol burnt = 1.379 g
Initial temperature of water = 17.7°C
Maximum temperature of water = 42.5°C

a. Explain why the temperature of the water increased between 100 and 270 seconds. (1 mark)

* exothermic reaction

OR * heat being transferred into container.

b. Explain why the temperature gradually decreased between 300 and 500 seconds. (1 mark)

* heat being transferred (lost) to surroundings.
(imperfect insulation)

c. Would you expect the result (ΔH, *in kJ mol*$^{-1}$) to be similar or quite different if butan-1-ol was used instead? Briefly explain your reasoning. (2 marks)

* different/higher for butan-1-ol.

* (- larger molecule so more bonds broken/formed.
OR
- 1 mol butanol has much higher mass (ie more fuel).
and similar kJ/g.)

d. Write a concise conclusion for this student's experiment. (2 marks)

* Explicit reference to AIM (½) and RESULT (½)

* Result supported by key DATA

END OF SAC